Tutorial Lab 12.3 (Formatively Assessed):

Large Signal and Small Signal Operation of FETs

Preliminary Matter: Setting up the Cadence Environment

Please read through sections 1 through 3 of the tutorial lab 12.1 instruction booklet/handout before entering these commands into your terminal.

For this course, you will use the Cadence Virtuoso Design Environment (referred to simply as Cadence). Cadence is a Linux based software suite for designing full custom integrated circuits. In order to use Cadence you will first need to set-up the Cadence environment.

8. Open a terminal window and ensure that you are in your home directory using the following command:

```
cd ~
```

Note: 'cd' stands for change directory. The solitary tilde symbol (~) is a shortcut/alias for your home directory (/home/username).

9. Under your home directory, use the following commands to load the Cadence environment:

```
module use /eda/cadence/modules
module load course/EENGM6030
```

Note: The two 'module' commands load the required shell/Cadence environmental variables for this course (paths, names, aliases…) In particular, this includes a reference to the process design kit (PDK) used in this course, the GPDK180. You will need to do this every time you open a terminal, since the sourced environment only applies to the current terminal (until it's closed).

a. ***Note: This step only needs to be completed ONCE - If you already ran this script in lab 12.1, you don't need to run it again here.*** The first time you use Cadence, you will need to run the following command (script):

```
EENGM6030_setup
```

This will create a folder called AICD (short for Analogue Integrated Circuit Design) under your home directory. The AICD folder contains the files and directories needed to run Cadence. Running this setup script more than once will result in a warning message that the AICD folder already exists. It will then exit without making changes.

10. The AICD folder will be used for all your Cadence related activities for this course. **You should always make sure you are in your AICD folder before you launch Cadence.** To run cadence, use the following commands:

```
cd ~/AICD
virtuoso &
```

Note: The 'virtuoso &' command starts Cadence. The ampersand symbol (&) causes the Cadence process to run in the background freeing up the terminal for other commands the '&' symbol is a feature of the terminal's shell (bash) not Cadence.

1. Learning Objectives of this Lab:

From the previous lab, you should know how to:

- Parameterise components in a schematic using design variables. Copy design variables from the schematic into the ADE. Initialise the design variables in the ADE.

- Carry out a DC operating point analysis. Annotate a schematic with DC node voltages, DC operating point information, and other model/circuit parameters.

- Carry out a DC sweep analysis and a parametric analysis (using design variables). Use the parametric analysis tool to add more variables to a DC-sweep.

- Simulate the terminal characteristics of the nmoshv/pmoshv FETs from the GPDK180 technology library. Plot the I_{DS} vs. V_{GS}, I_{DS} vs. V_{DS}, and I_{DS} vs. V_{DS} vs. V_{GS} family of curves.

- Create and simulate a resistor-loaded NMOS FET single-stage amplifier. Understand the conflicting requirements of the simple resistor-loaded FET amplifier – specifically, the need to maximise the small-signal gain of the amplifier by using a large load resistance but at the same time the need to constrain the size of the load resistance in order to set an appropriate DC operating point.

- Create and simulate a PMOS current source loaded NMOS FET single-stage amplifier. Understand how the use of a current source load allows us to simultaneously and independently maximise the small-signal gain of the amplifier (infinite small-signal resistance) and set the DC operating point of the circuit at an appropriate level (i.e., that its DC current is independent of its small-signal resistance).

In this lab, you will:

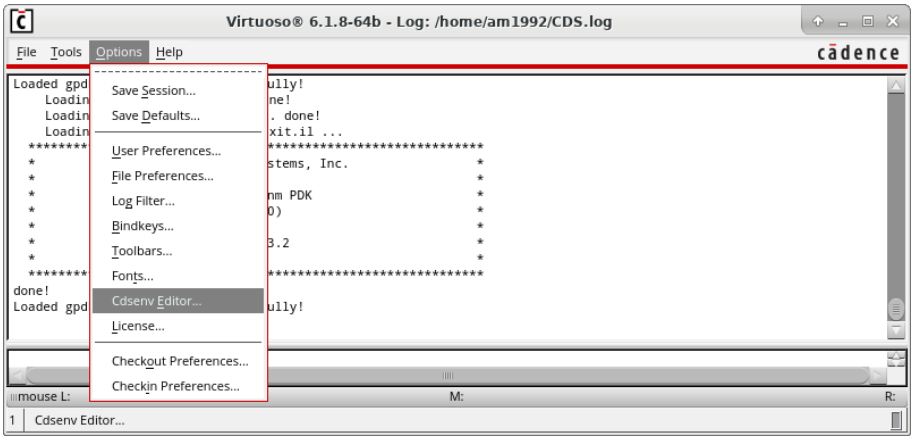
- Develop the small-signal model of the FET amplifier from the large-signal terminal characteristics simulated in the previous lab. Determine how to set the bias point to maximise performance. Understand the role of operating point calculations in AC analysis. Carry out a frequency sweep analysis of a resistor-loaded NMOS FET single-stage amplifier using an AC analysis.

2. Cadence Documentation, Help, and Licensing

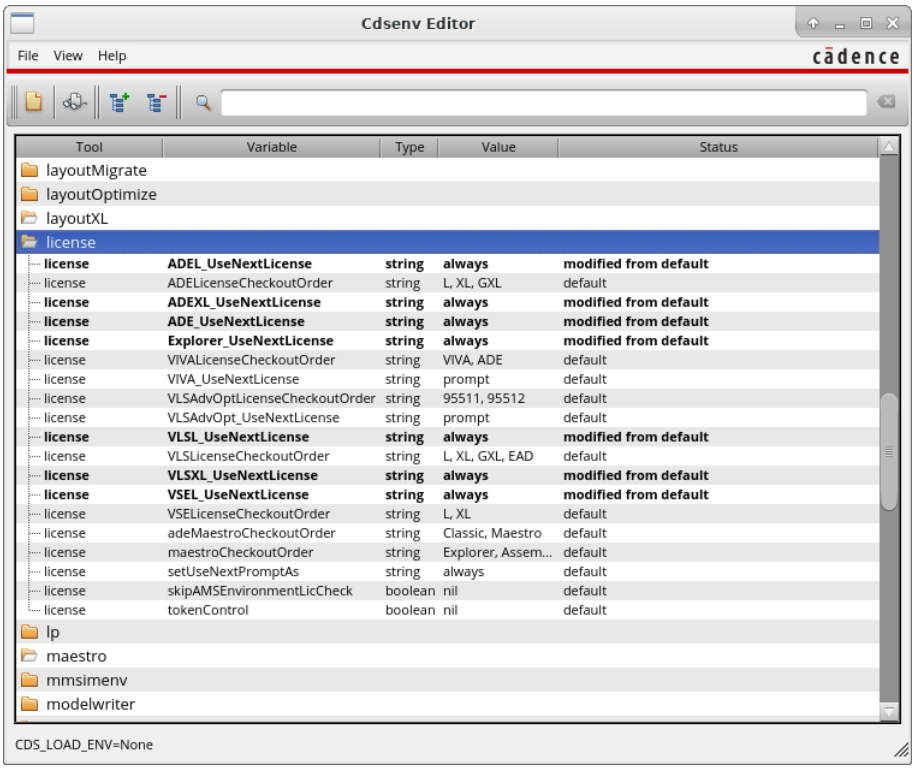
Local and online help is available from the help menus (**Menu: Help → User Guide** or by pressing **F1**), or by typing 'cdnshelp &' at the command prompt (assuming you have setup your terminal environment).

PLEASE NOTE: WHEN LAUNCHING A TOOL IN THE CADENCE DESIGN ENVIRONMENT YOU MAY SOMETIMES GET A WARNING ABOUT A LICENSE BEING UNAVAILABLE FOR THE TOOL, AND REQUESTING PERMISSION TO OBTAIN A HIGHER-TIERED LICENSE. YOU SHOULD SELECT 'ALWAYS' TO THESE MESSAGES.

If you accidently deny a license request, or you cannot access a license, you will need to modify the Cadence environment licence/permissions from the CIW. From the CIW menu go to **Menu: Options → Cdsenv Editor**. Once in the environment editor, go to the license folder, and make sure all licenses (in bold) are set to '**always'** – if any entries are listed as '**never**' you will need to manually change/retype them to '**always**'.



< Figure X.1. The Cadence Environment Menu Option from the CIW Window. >



< Figure X.2. The Cadence Environment Editor. Note, all licenses should be set to 'always'.>

3. Getting Started with Cadence:

For a refresher/introduction on how to start a Cadence session please read through sections 1 through 3 of the tutorial lab 12.1 instruction booklet/handout.

3.1 Dealing with Lock files

PLEASE NOTE: IF A SCHEMATIC ISN'T PROPERLY CLOSED DOWN (FOR EXAMPLE, DUE TO A DROPPED REMOTE CONNECTION), THE SCHEMATIC MIGHT BECOME EDIT LOCKED. IF THIS HAPPENS YOU WILL NOT BE ABLE TO OPEN THE FILE FOR EDITING. YOU WILL ONLY BE ABLE TO OPEN THE FILE FOR READING UNTIL YOU REMOVE ANY/ALL LOCK FILES.

There are two ways you can unlock files:

(3) The first is to manually delete any edit lock files. To do this, go to your '~/AICD/Designs' folder and find the folder representing the schematic e.g., for lab1 this would be '~/AICD/Designs/lab121_vdivider/schematic' if this is what you called your schematic. Delete all the .cdslck lock files in this folder and you should be able to open your schematic for editing.

(4) You can also automatically release edit locks with the command:

```
clsAdminTool -are .
```

To use this command, you have to be in your AICD folder (the full-stop is shorthand for the current directory path). If you are not in your AICD folder, you have to specify the path (relative or absolute) to your AICD directory.

Since this is a Cadence command, you have to ensure that you have set up the terminal environment for Cadence (in particular, the modules must be loaded as in step 2, 'Preliminary Matter'). It is best to use the terminal window that you used to start Cadence, since the environment was already set up. Note that you will not be able enter any commands into your terminal if you did not start cadence with the '&' character. By example, to release any lock files after opening a new terminal, the full command set would be,

```
cd~
module use /eda/cadence/modules
module load course/EENGM6030_2021
clsAdminTool -are AICD
```

For more information on this command, use

```
clsAdminTool -help
```

4. Circuit Simulation Exercises: Part 1

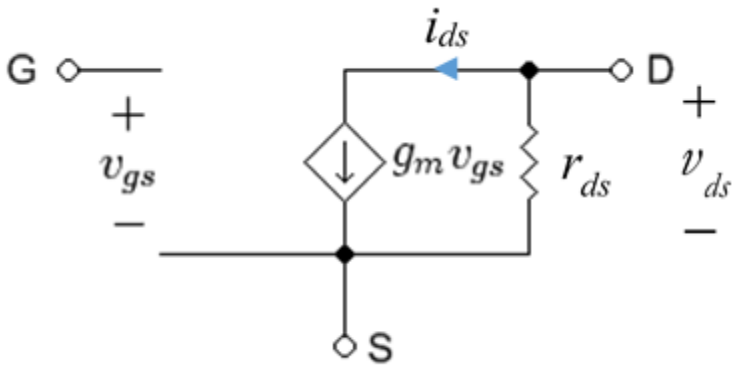
With guidance, you will:

- Examine the relationship between the overdrive voltage and the transconductance. Understand weak channel-inversion, strong channel-inversion and carrier-velocity saturation.

- Understand the small-signal model of a FET amplifier at a bias point. Determine how best to choose a bias point given the large-signal terminal characteristics of an NMOS/PMOS device.

4.1 The Small-Signal Model

Consider, the simplest small-signal model of the MOSFET,



< Figure 58. >

Where g_m is the small-signal transconductance, the proportionality constant that relates the output signal current i_{ds} to the input signal voltage v_{gs}, and where the small-signal resistance r_{ds} models the non-ideal dependency of i_{ds} on the output signal voltage v_{ds}. In this lab, we will relate the large-signal characteristics of the NMOS device to the small-signal model and determine – through simulation – how to choose the DC operating point. We will revisit the small-signal model (Figure 1) in part 2 of this lab when we introduce the AC analysis.

4.2 Plotting the Small-Signal Transconductance and Output Impedance Curves

(4.2.1) In the large-signal domain, I_{DS} is a non-linear function of V_{GS} and V_{DS},

$$I_{DS} = f(V_{GS}, V_{DS})$$

The small-signal transconductance g_m and output impedance r_{ds} are defined as,

$$g_m(V_{GS}) = \frac{\partial I_{DS}}{\partial V_{GS}} \qquad r_{ds}(V_{DS}) = \left[\frac{\partial I_{DS}}{\partial V_{DS}}\right]^{-1}$$

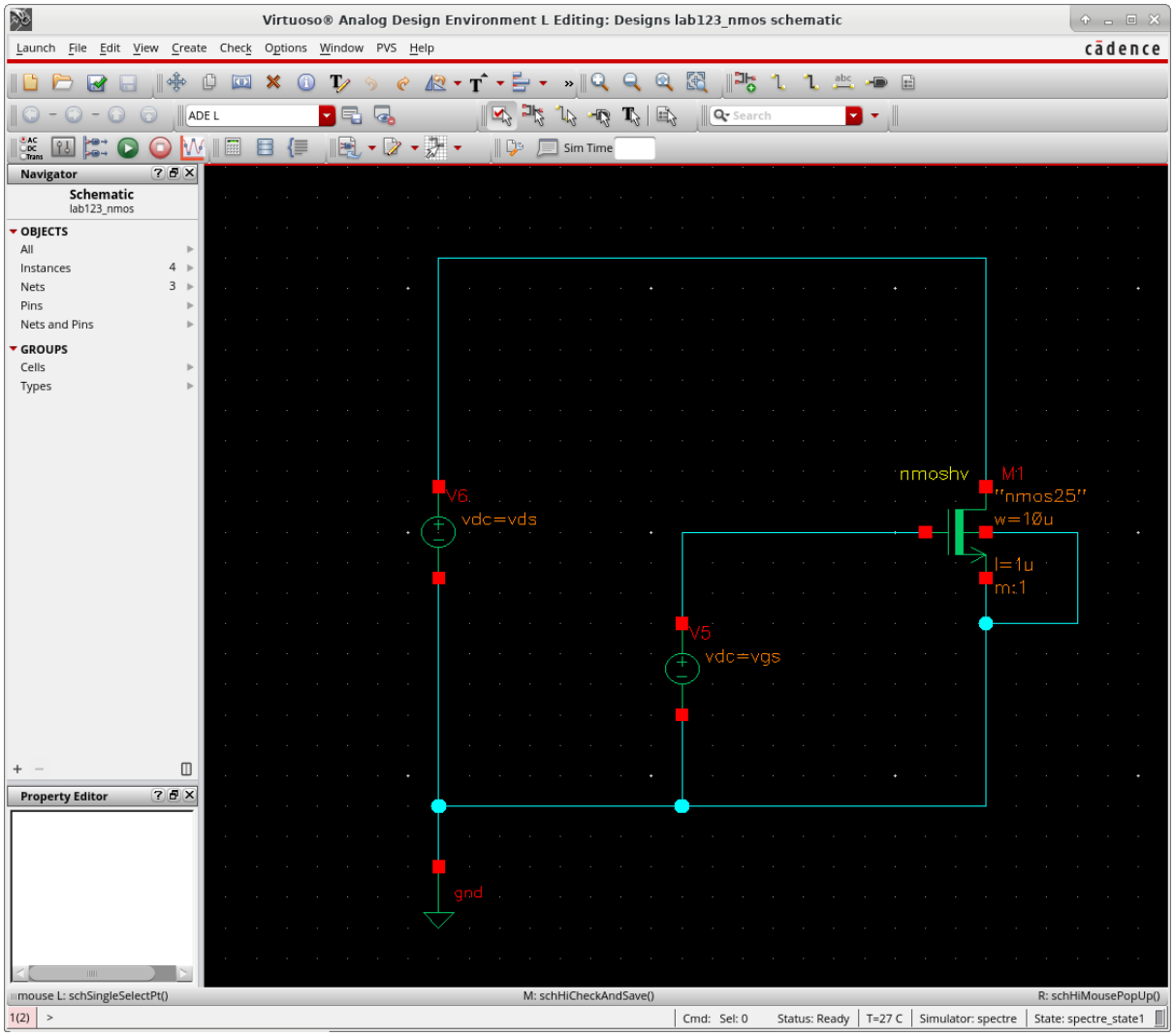
The derivative of the large-signal I_{DS} vs. V_{GS} plot will therefore give us a plot of g_m vs. V_{GS}. Likewise, the derivative of the large-signal I_{DS} vs. V_{DS} plot will give us a plot of r_{ds} vs. V_{DS}. For a fixed bias point, the small-signal parameters g_m and r_{ds}, are simply a single point along their respective curves. (Note that in the case of g_m, we assume V_{DS} is held constant and in the case of r_{ds} we assume V_{GS} is held constant.)

(4.2.2) In part 1 of the previous lab [12.2], we plotted the curves of I_{DS} vs. V_{GS} (with V_{DS}=2.5V) and I_{DS} vs. V_{DS} (with V_{GS}=0.7V) for an unloaded NMOS FET. From these graphs we want to obtain the curves of g_m vs. V_{GS} and r_{ds} vs. V_{DS}. Cadence virtuoso has an in-built calculator that allows us to post-process the results of our simulations. We re-run last lab's simulations and use the cadence calculator to obtain the required curves.

(4.2.3) Open the schematic you created in part 1 of lab 12.2 ('lab122_nmos' or whatever name you chose). You can also copy the cell to create a new file e.g. lab123_nmos. (You should now know how to do this without instruction.)

(4.2.4) Ensure that your schematic is properly parameterised using design variables: recall, for the DC supply voltage vdc=vds and for the DC gate-to-source voltage vdc=vgs. The NMOS device should be W/L = 10u/1u. See Figure 2. Ensure that it checks and saves without warnings or errors. Use the CIW to diagnose any warnings or errors.

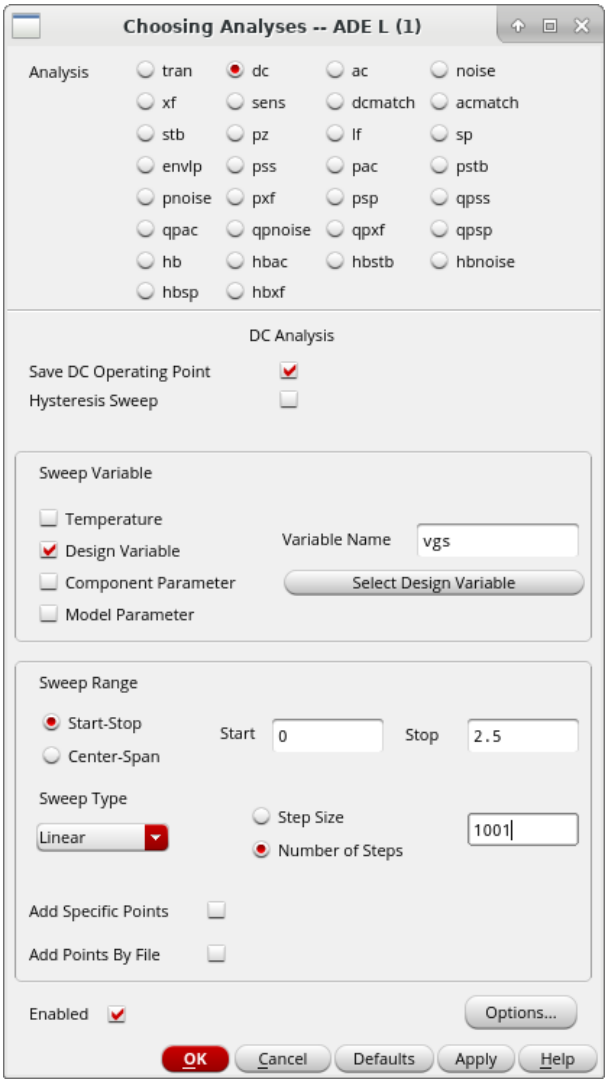
(4.2.5) From the schematic window, launch the ADE tool (**Menu: Launch à ADE L**). From the ADE window, copy the design variables from the schematic into the ADE (**Menu: Variables à Copy From Cellview**). You should now see the *vds* and *vgs* variables in the 'Design Variables' list of the ADE. Initialise *vds = 2.5* and *vgs = 0.7*. Note, that these initial values are only used to calculate the DC operating point of the circuit.



< Figure 59. The NMOS (10u/1u) testbench circuit with Design Variables for the DC supply voltage (vdc=vds) and the DC gate-to-source voltage (vdc=vgs). >

(4.2.6) From the ADE window, setup the DC sweep analysis for I_{DS} vs. V_{GS} (**Menu: Analyses à Choose**). We wish to sweep the input voltage (vgs) of the NMOS device from 0V to 2.5V while observing the drain current (IDS). We will use 1001 steps in the sweep to give the output waveform sufficient resolution.

In the choosing analyses form, select the '**dc**' option button. Check the 'Save DC Operating Point' box. Check the 'Design Variable' box as the sweep variable. Type *vgs* as the variable name. Choose the 'Start-Stop' option button with Start = 0 and Stop = 2.5 for the sweep range. Select 'Linear' from the drop down menu as the sweep type. Choose the 'Number of Steps' option button with 1001 steps in the sweep. Ensure that the analysis is enabled by checking the 'Enabled' box near the bottom of the form. See Figure 3. OK the form. The **dc** analysis should appear in the 'Analyses' list of the ADE window.



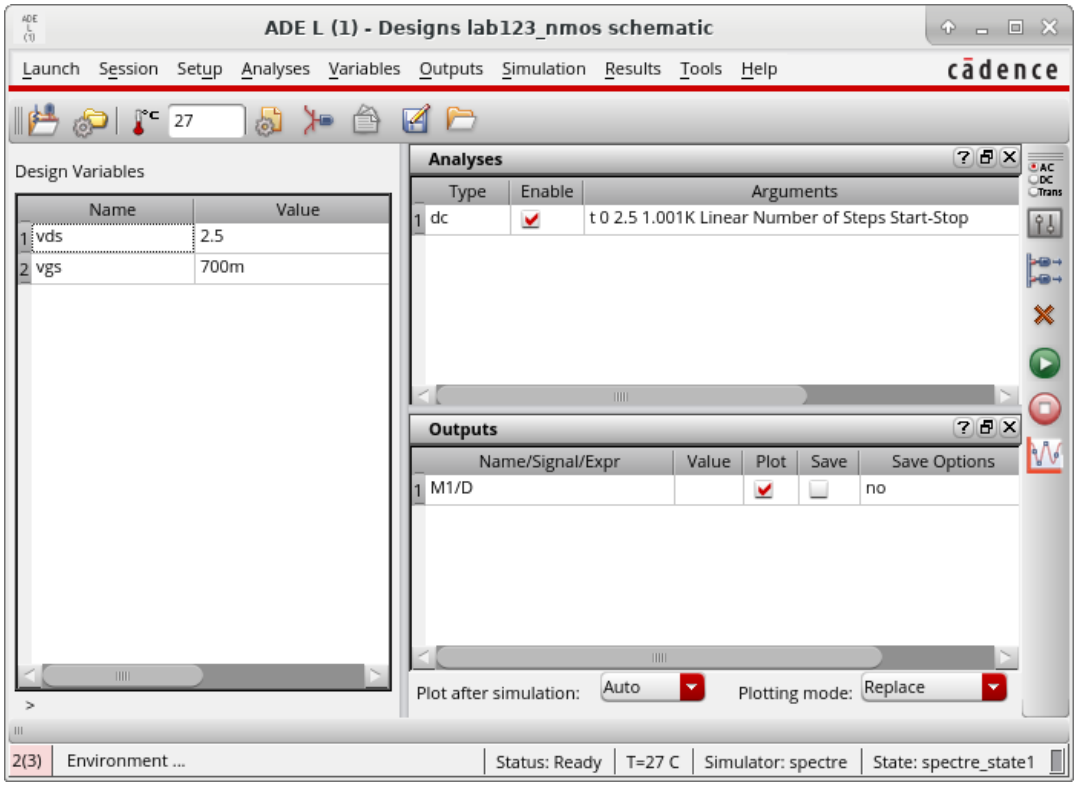
< Figure 60. The Choosing Analyses Form for a DC sweep. >

(4.2.7) Select only the drain node current (I_{DS}) for plotting. From the ADE, select **Menu: Outputs à To Be Plotted à Select On Schematic** to enter the plotting mode of the ADE**.** Select only the drain node of the NMOS device (M1/D). **Press the 'ESC' key to exit the plotting mode of the ADE**.

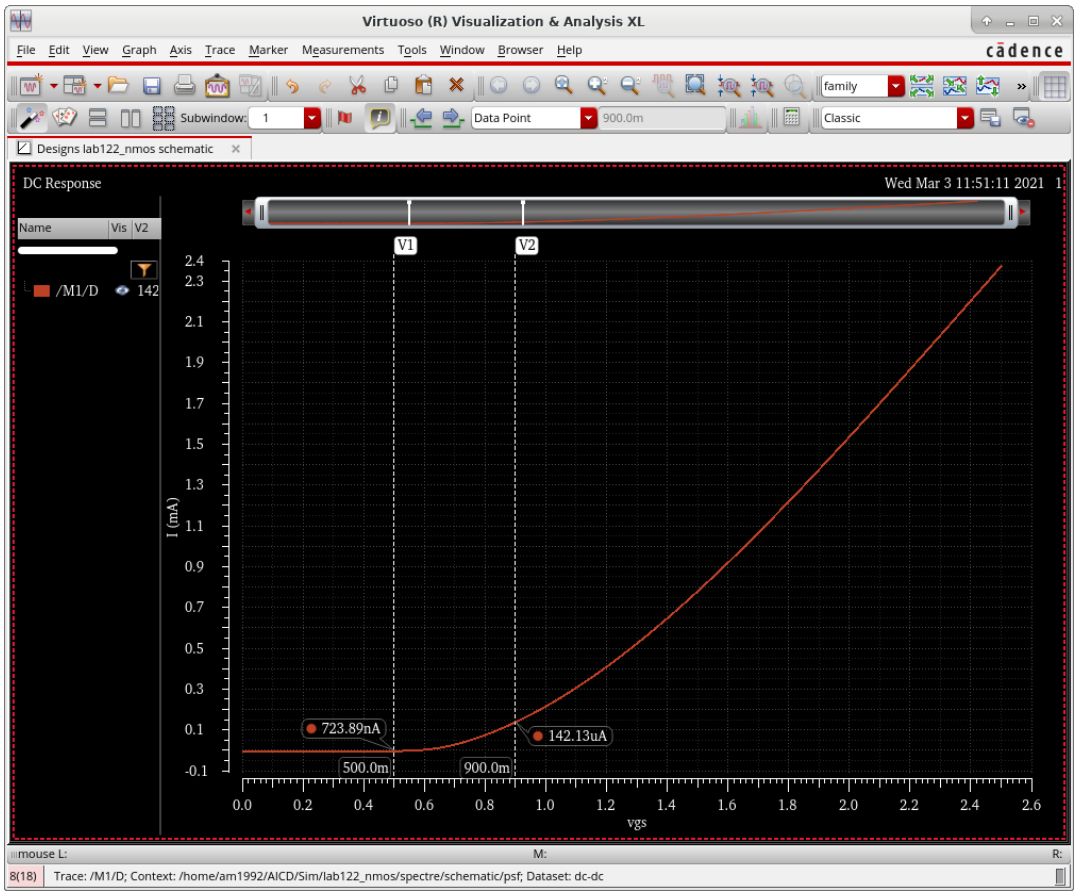
*Recall**,** if you want to plot a particular voltage, you click on the net/wire. If you want to save and plot the currents, you have to click a terminal **node** of a component (not the net/wire).*

(4.2.8) In the ADE window, run the '**Netlist and Run**' command by clicking on the green play button. The simulation should run without error and both the text output window as well as the waveform output window should open automatically. You should get a plot of I_{DS} vs. V_{GS}. See Figure 5.

*Recall, you can add point, horizontal, and/or vertical markers to your graph. From the waveform output window, choose **Menu: Markers à Create Marker…** to bring up the 'Create Graph Marker' form. Or use the keyboard shortcuts to add a marker to your plot: '**m**' (point marker), '**h**' (horizontal marker), and '**v**' (vertical marker).*



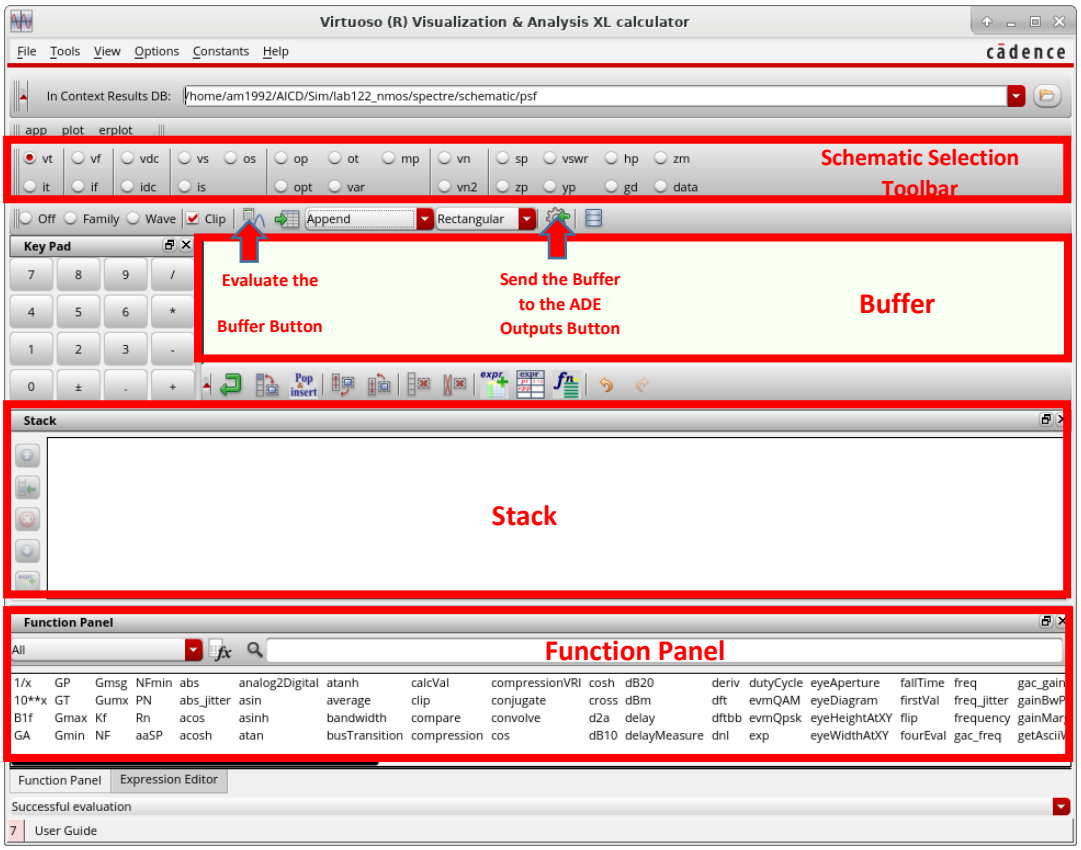
< Figure 61. The ADE with the Design Variables, Analyses and Outputs. >



< Figure 62. The waveform output of IDS vs. VGS with vertical markers at 0.5V and 0.9V. >

4.3 Using the Built-in Waveform Calculator

(4.3.1) The calculator tool allows you to post-process your results by applying mathematical operations to your simulation data. To launch the calculator tool, from the ADE window, choose **Menu: Tools à Calculator…**



< Figure 63. The VIVA XL Calculator. >

(4.3.2) The principal components of the calculator are:

1. **Schematic Selection Toolbar.** Near the top of the calculator is the Schematic Selection Toolbar. This is how we get data from the simulation into the calculator. This is similar to the 'To Be Plotted' command in the ADE. This toolbar contains a selection of option/radio buttons for selecting the nets (voltages) and/or nodes (currents) on the schematic following a particular analysis. When you select one of the option buttons, the schematic will come into focus allowing you to select the required net or node.

2. **Buffer**. When you select the net or node using the schematic selection toolbar, or by sending a trace from the waveform output window directly to the calculator, the expression underlying the net, node, or waveform will appear in the buffer of the calculator. Any mathematical operations performed on the expression will also be shown in the buffer.

3. **Stack**. You can push/store any expression you have built in the buffer on to the stack by either pressing enter on the keyboard or using the enter icon button (just beneath the buffer area). You can also load/pop an expression (assuming you have any stored) from the stack into the buffer. You can also swap an expression on the stack with the expression in the buffer. The stack can be up to eight instructions deep.

4. **Function Panel**. The function panel contains a large number of standard mathematical and engineering functions that can be applied to the expression in the buffer. Expressions of any complexity can be built, using as many functions as required. The function you are applying has to be compatible with the data from the simulation. The list of functions are categorised according to the drop down menu in the function panel. Take some time to explore the function list/categories.

5. **Evaluate the Buffer Button**. This button evaluates the expression in the buffer. Depending on the expression, and the mathematical functions applied to the expression, the buffer may evaluate as a scalar or as a waveform. A scalar evaluation will be displayed in the buffer area – replacing the original expression. A waveform evaluation will be displayed in a waveform output window. The waveform will be plotted according to the calculator plotting mode in the drop down menu next to the evaluate button (i.e. replace, append, new Win, new SubWin). Append is the default.

6. **Send Buffer Expression to ADE Outputs**. Once an expression has been built in the buffer of the calculator, it can be sent to the Outputs section of the ADE. This expression will then be treated the same as any other output in the list.

(4.4.3) Note: The calculator has two input modes: Reverse Polish Notation (RPN) and Algebraic mode. The calculator mode is set to RPN by default. Change this to algebraic. From the calculator window choose **Menu: Options à Mode à Algebraic Mode**. This setting doesn't change how the calculator operates, but it does change the behaviour of the calculator when adding or manipulating expressions to/in the buffer. The most important behaviours to note are (1) in algebraic mode, adding a function (from the function panel) to the buffer will place that function wherever the cursor is located, and (2) in algebraic mode, when applying a function you need to manually add the closing bracket.

(4.3.4) Note: An expression in the calculator will only successfully evaluate if the simulation data is compatible with the expression in the buffer. When you use the Schematic Selection Toolbar, each button can only be used to acquire the simulation data for a specific analysis type. The following rules apply:

vt – select transient analysis data at a net (voltage).
it – select transient analysis data at a node (current).

vf – select AC 'frequency response' analysis data at a net (voltage).
if – select AC 'frequency response' analysis data at a node (current).

vdc – select DC operating point analysis data at a net (voltage).
idc – select DC operating point analysis data at a node (current).
vs – select DC sweep analysis data at a net (voltage).
is – select DC sweep analysis data at a node (current).
os – select DC sweep analysis data (operating point information) of a device.

It is possible to build up an expression using the wrong option buttons. For example, if you use **vt** (transient data) when you have only carried out a DC analysis, you will get an error on evaluation ('`Expression evaluates to nil`'). So, make sure you use the correct schematic selection button for the simulation. The other option buttons are associated with analyses we do not use in this course. Feel free to peruse the user guide for more information (**Menu: Help à VIVA XL User Guide**)

(4.3.5) Note: You can also send a signal to the calculator directly from the waveform output window (assuming you have plotted any waveforms). This is often the easiest method. Right-click on the trace in the waveform output window, and select **Send To à Calculator** from the context menu. This will send the underlying expression for the trace to the buffer of the calculator.

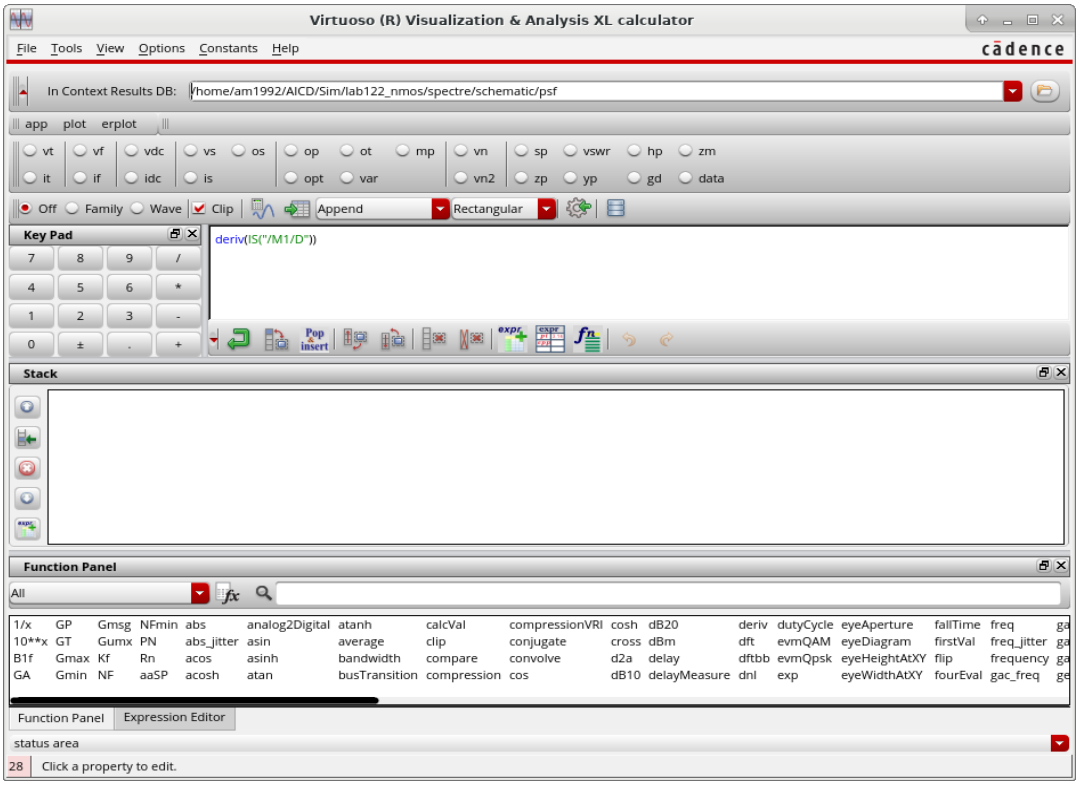
Plotting g_m vs. V_{GS}

(4.3.6) Use the schematic Selection Toolbar to get the plot of I_{DS} vs. V_{GS} into the calculator. Select the '**is**' option button (for current sweep) on the schematic selection toolbar. When you select the '**is**' button the schematic editor window will come into focus. Select only the drain node current (I_{DS}) by clicking on the drain node of the NMOS device (M1/D). The calculator should return to focus and you should see the expression for the drain current IS("/M1/D") in the buffer area of the calculator.

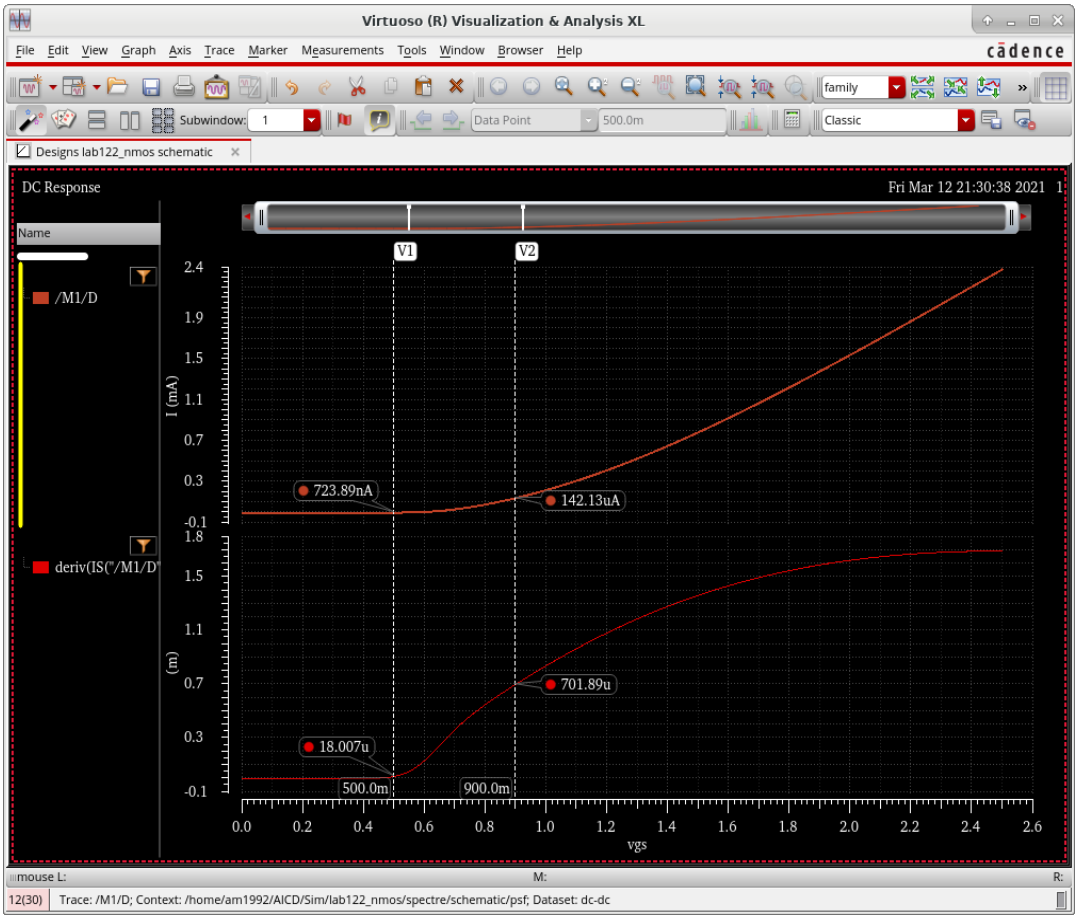
(4.3.7) With the expression for I_{DS} vs. V_{GS} in the buffer, apply the derivative function from the function panel. In the function panel of the calculator window, select '**All**' from the drop down menu. Place the cursor in the correct positon (or highlight the entire expression) in the buffer and select the '**deriv**' function from the list of functions. The functions are listed alphabetically so either search the list or use the text search box just above the listing area. (Note that the **deriv** function is categorised under special functions.) Make sure to close the brackets after you select the **deriv** function. The expression in the buffer should be deriv(IS("/M1/D")). See Figure 7.

(4.3.8) In the calculator window, evaluate the buffer expression by using the **Evaluate the Buffer** button. If you set up the expression correctly and the simulation data exists (you should have run the DC sweep simulation before opening the calculator) the waveform of dI_{DS}/dV_{GS} vs. V_{GS} should be plotted alongside the original plot (assuming you didn't close it down and that the calculator plotting mode). This is a plot of the small-signal transconductance of the device versus the DC gate-to-source bias voltage - g_m vs. V_{GS}.

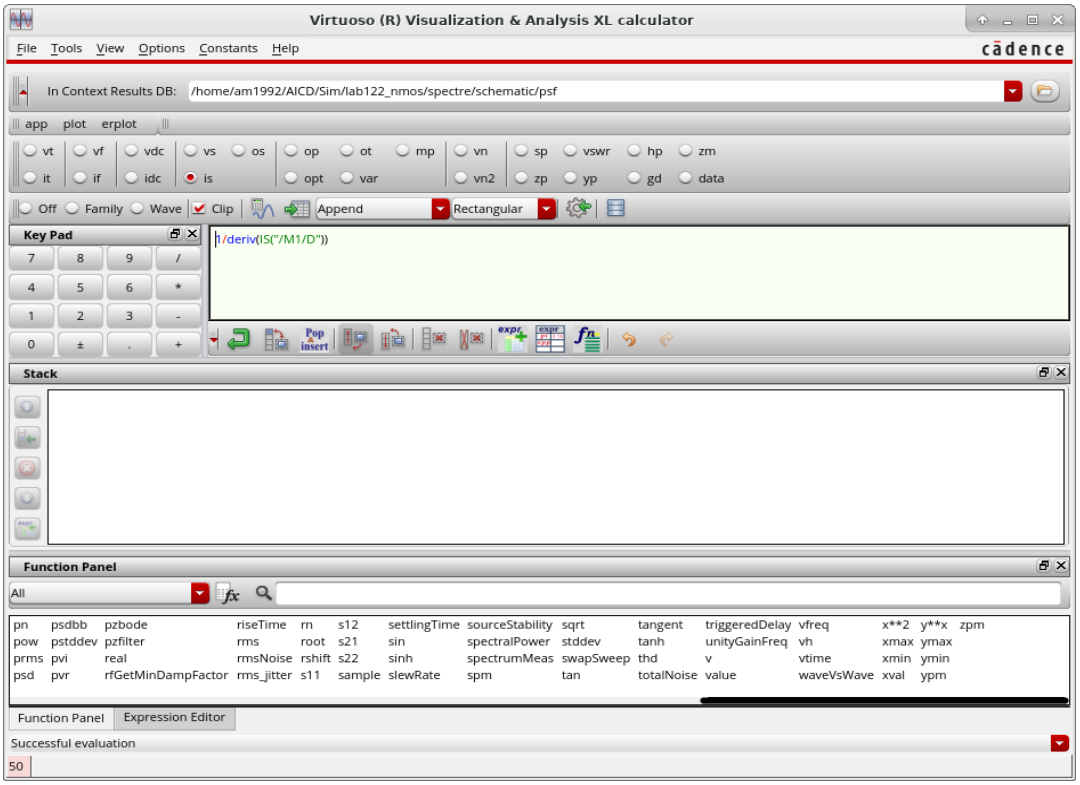
(4.3.9) Split the traces for clarity. Right-click on the empty area of the graph window and select S**plit Current Strip à trace** from the context menu.



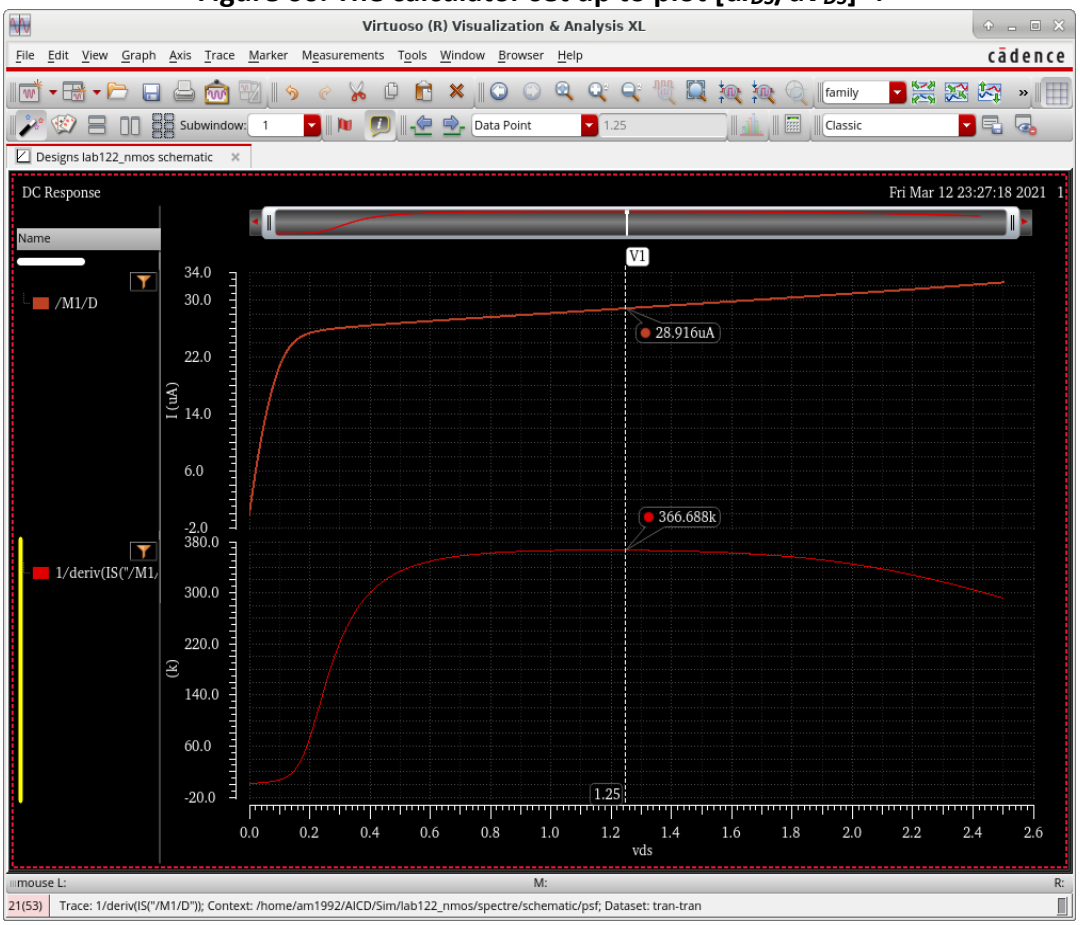
< Figure 64. The calculator set up to plot dI_{DS}/dV_{GS}. >



< Figure 65. The plots of I_{DS} vs. V_{GS} (top) and g_m=dI_{DS}/dV_{GS} vs. V_{GS} (bottom). V_{DS}=2.5. >



< Figure 66. The calculator set up to plot [dI_{DS}/dV_{DS}]$^{-1}$. >



< Figure 67. The plots of I_{DS} vs. V_{DS} (top) and r_{ds}=dI_{DS}/dV_{DS} vs. V_{DS} (bottom). V_{GS}=0.7. >

Plotting r_{ds} vs. V_{DS}

(4.3.10) Plot I_{DS} vs. V_{DS}. From the ADE window, edit the DC analysis (**dc**) to make V_{DS} the sweep variable. Either double click on the analysis or right-click on the analysis and choose edit from the context menu. See Figure 3. Change the sweep variable name from V_{GS} to V_{DS}. Leave the rest of the form unchanged. Make sure V_{GS} = 0.7V in the ADE. OK the form.

(4.3.11) In the ADE window, run the DC sweep simulation. The simulation should run properly and both the text output window as well as the waveform output window should open automatically.

(4.3.12) The expression from the previous exercise should still be in the buffer. If not, use the schematic selection toolbar to get the plot of I_{DS} vs. V_{DS} into the calculator following the same steps as above. To get the reciprocal of the expression you can manually modify the expression in the buffer to either 1/deriv(IS("/M1/D")).

(4.3.13) In the calculator window, evaluate the buffer expression by using the **Evaluate the Buffer** button. If you set up the expression correctly and the simulation data exists (you should have run the DC sweep simulation before opening the calculator) the waveform of dV_{DS}/dI_{DS} vs. V_{DS} should be plotted alongside the original plot (assuming you didn't close it down and that the calculator plotting mode). This is a plot of the small-signal output resistance of the device versus the DC drain-to-source bias voltage - r_{ds} vs. V_{DS}. At a value of $V_{DS} = V_{dd}/2$=1.25V, the output resistance is r_{ds} = 366.688KΩ (Figure 10, bottom).

4.4 Discussion: Choosing the Bias Point

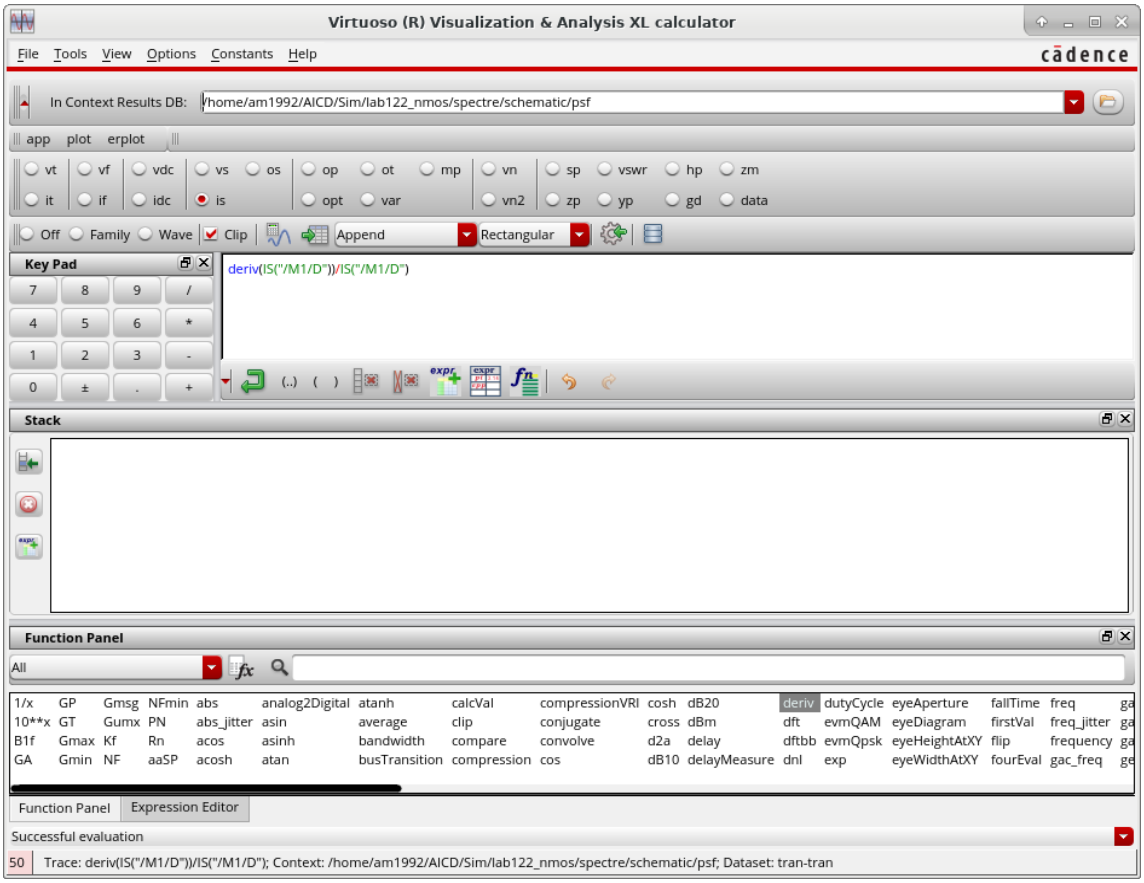
We are not just interested in the value of g_m. We also need to consider the power draw for a given g_m. As we increase the drain current, and consequently power consumption, we get a diminishing return on g_m; that is, we spend a lot of power for only a small increase in g_m. The metric g_m/I_{DS} is measure of bias efficiency – it tells us where on the curve of I_{DS} vs. V_{GS} we are getting the most return (g_m, gain) for our current draw (power consumption).

Plotting g_m/I_{DS} vs. V_{GS}

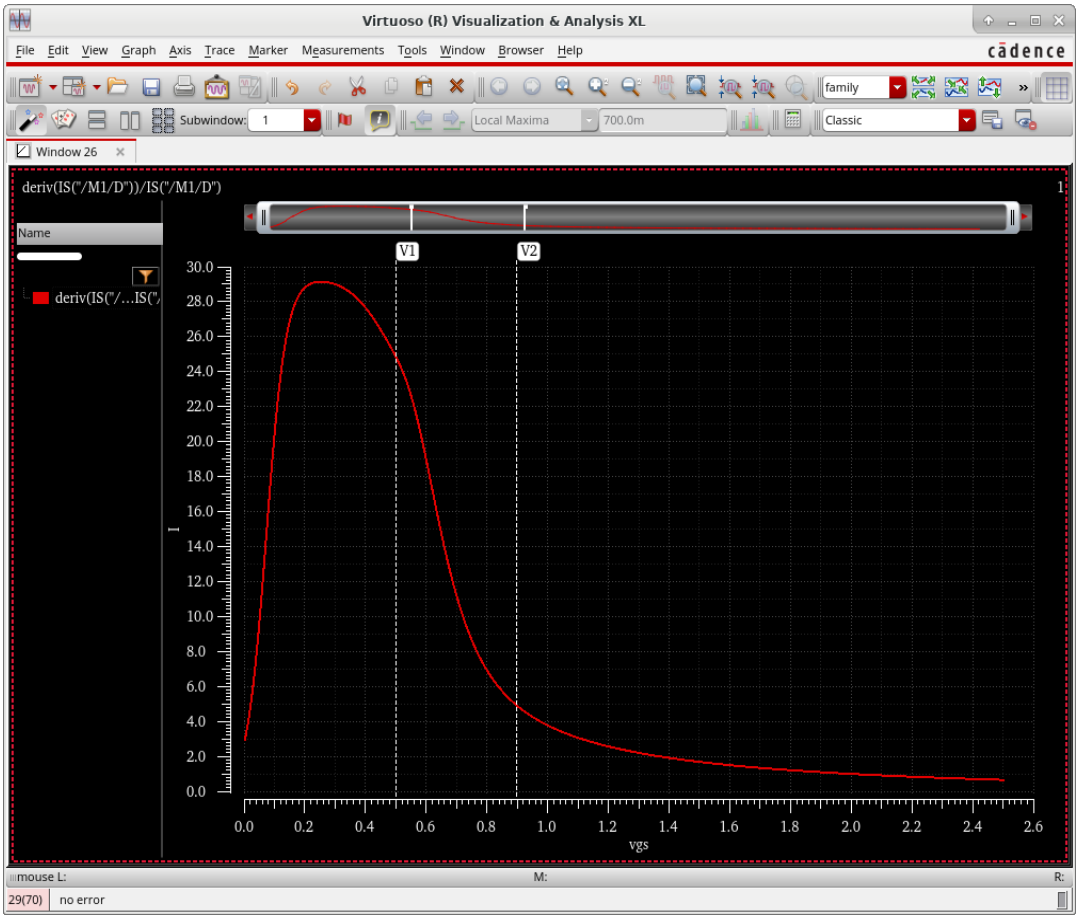
(4.4.2) We will plot the curve of g_m/I_{DS} vs. V_{GS}. using the calculator. Change the sweep variable back to V_{GS} and re-simulate the DC sweep analysis (see Figure 3). As you did above, build the expression deriv(IS("/M1/D"))/(IS("/M1/D")) in the calculator buffer. This will plot g_m/I_{DS} vs. V_{GS}. See Figure 11 and 12.

Use the schematic Selection Toolbar. Select the 'is' option button (for DC current sweep) on the schematic selection toolbar. When you select the 'is' button the schematic editor window will come into focus. Select only the drain node current (I_{DS}) by clicking on the drain node of the NMOS device (M1/D). The calculator should return to focus, and you should see the expression for the drain current IS("/M1/D") in the buffer area of the calculator. Use the ***deriv*** *function to plot the derivative of the drain current deriv(IS("/M1/D")). This will give us a plot*

*of g_m vs. V_{GS}. Add a backslash / character to the end of the expression for division. To add the final current term, you can either copy/paste the string to the end of your expression or use the schematic selection tool to append the drain current to the expression by selecting the '**is**' button and again clicking on the drain node of the NMOS device (M1/D). You should now have deriv(IS("/M1/D"))/(IS("/M1/D")) in the expression buffer. See Figure 11.*



< Figure 68. The calculator set up to plot the derivative of the g_m/I_{DS} vs. V_{GS}. >



< Figure 69. The plot of g_m/I_{DS} vs. V_{GS}. V_{DS}=2.5. >

*final current term, you can either copy/paste the string to the end of your expression or use the schematic selection tool to append the drain current to the expression by selecting the '**is**' button and again clicking on the drain node of the NMOS device (M1/D). You should now have deriv(IS("/M1/D"))/(IS("/M1/D")) in the expression buffer. See Figure 11.*

(4.4.3) **Discussion**. In section 1.2.5 (page 11) referring to fig. 3 of the background material, note that there are four regions of the I_{DS} vs. V_{GS} characteristic curve – the sub-threshold region, the exponential region, the square-law region and the velocity saturation region.

When biasing a MOSFET for use as an amplifier, we want to bias the device in its square-law region of operation, where the drain current is proportional to the square of the overdrive voltage. In this region, we have a strong channel-inversion giving us a good return in terms of a performance vs. power consumption trade-off. Recall the first-order equation for the drain current in the square-law region (assuming the device is in the saturation region) is,

$$i_{DS} = \frac{1}{2}k'\left(\frac{W}{L}\right)(V_{GS} - V_t)^2(1 + \lambda V_{DS})$$

where $k = k'\left(\frac{W}{L}\right) = \mu C_{ox}\left(\frac{W}{L}\right)$. There are three inter-related expressions for the small-signal transconductance g_m in the square-law region (repeated here).

$$g_m = k(V_{OV}) = 2\sqrt{ki_{DS}} = \frac{2i_{DS}}{V_{OV}}$$

In general, we see that the transconductance is proportional to the *k* process parameter, the drain current I_{DS}, and the overdrive voltage $V_{OV} = V_{GS} - V_t$. The higher the overdrive voltage or the higher the drain current, the higher the transconductance. This holds true up until we reach velocity saturation. In the velocity saturation region, where the maximum velocity of the charge carriers in the channel has been reached, we get increasingly small returns on g_m (and consequently gain) for any increase in overdrive voltage or drain current and therefore power consumption – lowering overall power efficiency.

Lower the drain current or overdrive voltage too much, and we come out of the predictable square-law region – where all the design equations/approximations work – and enter the exponential region. Here, we have weak channel-inversion, where we operate awkwardly close to the threshold voltage – where process, voltage, and/or temperature (PVT) variations may accidentally turn the MOS device off completely (i.e., we cross into the sub-threshold region). In addition, the exponential region of operation is noisy due to the low drain current density.

We find as a general rule, working with an overdrive voltage of 0.2V (VGS=0.7V) – we get a good transconductance for the bias current draw, g_m ~ 365uS. This puts us firmly in the square-law/strong-inversion region, sufficiently away from the weak-inversion/exponential region where noise, PVT, and model accuracy are problematic, and away from the velocity saturation region where power efficiency is poor. Moreover, aiming for a V_{OV}=0.2V holds true regardless of process node or technology.

What about VDS? In the equation for IDS above, the output resistance is determined by the channel length modulation parameter λ. From the background material, the MOSFET output resistance is then given by,

$$r_{ds} = \frac{1}{i_{DS}\lambda}$$

Recall from the previous lab that we choose VDS≈VDD/2 to give us maximum signal swing. We note from Figure 10, that the small-signal resistance r_{ds} is approximately flat over the range 0.6V to 2V (with r_{ds} ≈ 350–370kΩ). If we wanted to maximise linearity, a VDD of approximately 1.2V would be optimal – which is close enough to mid-rail.

So, for the fixed bias point VGS=0.7V and VDS=1.2V, IDS ≈ 28.8uA, r_{ds} ≈ 367kΩ, and g_m ≈ 365uS. From the large-signal characteristics (Figures 8 and 10) we have estimated the small-signal parameters g_m and r_{ds} (as in Figure 1).

In the next part of the lab, we will turn our attention to the AC analysis (aka small-signal analysis, linear analysis, or linear network analysis).

5. Circuit Simulation Exercises: Part 2

With guidance, you will:

- Understand the relationship between the AC analysis and the DC operating point.
- Simulate the frequency response of the basic resistor-loaded single-stage NMOS FET amplifier.

The AC analysis is a small-signal analysis. As we stated in the introduction (part 1), small-signal analysis assumes that a circuit has been linearised around a fixed DC operating point (or bias point). When you simulate a device using an AC analysis, you are not simulating the actual device but an idealised, linear model of the device. The input-output relationship of a circuit in an AC analysis is therefore *always* linear – regardless of the input signal size. (Note, that in an AC analysis, you can drive a circuit with any size of input – you are not limited to small perturbations of the input since you are not driving the actual circuit, but the linearised version of it. So, take care when interpreting the response of the simulation - just because you drive it with a 1V signal in AC analysis and get a linear response, this does not mean that it is realistic to do so with the actual circuit – e.g., due to clipping or distortion.)

When you carry out an AC (small-signal) analysis, the simulator first calculates the DC operating points. (Note that this happens automatically, there is no separate 'Save DC Operating Point' option, unlike the DC analysis.) The small-signal parameters of the devices/circuit are then calculated from the DC operating points. For a MOSFET, the small-signal parameters include voltages (vgs, vds, vgd…), currents (ids, igs, igd…), transconductances (gm, gmb…), parasitic capacitances (cgs, cds, cgd…), conductances (gds, gbd…), and many more. All of these parameters are used to construct the small-signal linear model of the MOSFET – which at its core is based on the circuit given in Figure 1 (although it has dozens of parameters). The linear input-output response of the device/circuit using the small-signal parameters is then calculated.

Aside: the full list of small-signal parameters for the GPDK180 NMOS is: *ids, isub, vgs, vds, vbs, vgb, vdb, vgd, vth, vdsat, vfbeff, gm, gds, gmbs, betaeff, cjd, cjs, qb, qg, qd, qbd, qbs, cgg, cgd, cgs, cgb, cdg, cdd, cds, cdb, csg, csd, css, csb, cbg, cbd, cbs, cbb, ron, id, is, ibulk, ibs, ibd, pwr, gmoverid, ueff, cgsovl, cgdovl, cgbovl, i1, i3, i4, gbd, gbs, vgsteff, qinv, igd, igs, igb, igcs, igcd, igidl, igisl, qgi, qsi, qdi, qbi, cddbi, cssbi, cggbi, cgsbi, cgdbi, cbdbi, cbsbi, qsrco, ide, ige, ise, ibe, idb, isb, vsb, gmb, vgt, vdss, vsat_marg, self_gain, rout, beff, fug, vearly, ft, vth_drive, vdsat_marg, tk, ib, ig.*

5.1 AC Analysis: DC Operating Point

(5.1.1) We will, again, plot the curve of g_m vs. V_{GS}. But, this time we will use an AC analysis to do so. Note that in this case, g_m will be calculated internally by the simulator, rather than by us explicitly as we did in part 1 (using the calculator). The point of this exercise is to demonstrate the underlying principle of the AC analysis (and its relationship to the DC operating point analysis).

(5.1.2) Recall, in the simplest small-signal model of the MOSFET, the drain current is given by,

$$i_{ds} = g_m v_{gs}$$

where the lowercase v_{gs}=1V is the AC input signal. If we sweep the DC bias V_{GS} (note the uppercase) using an AC analysis with the AC input signal set to 1V, the resulting plot of i_{ds} will also be a plot of g_m. Hence, by sweeping the DC bias point V_{GS} we will implicitly be plotting the curve of g_m vs. V_{GS} just as we did with the DC analysis.

(5.1.3) Note. We are using the same schematic and simulations as we used in part 1. If you don't have your schematic and ADE already open from the previous exercises. You will need to set up the schematic/ADE and ensure you have copied the design variables into the ADE and initialised them (see Figure 4). Also, make sure you have set up you DC sweep analysis with v_{gs} as your sweep variable (0-2.5, 1001, linear).

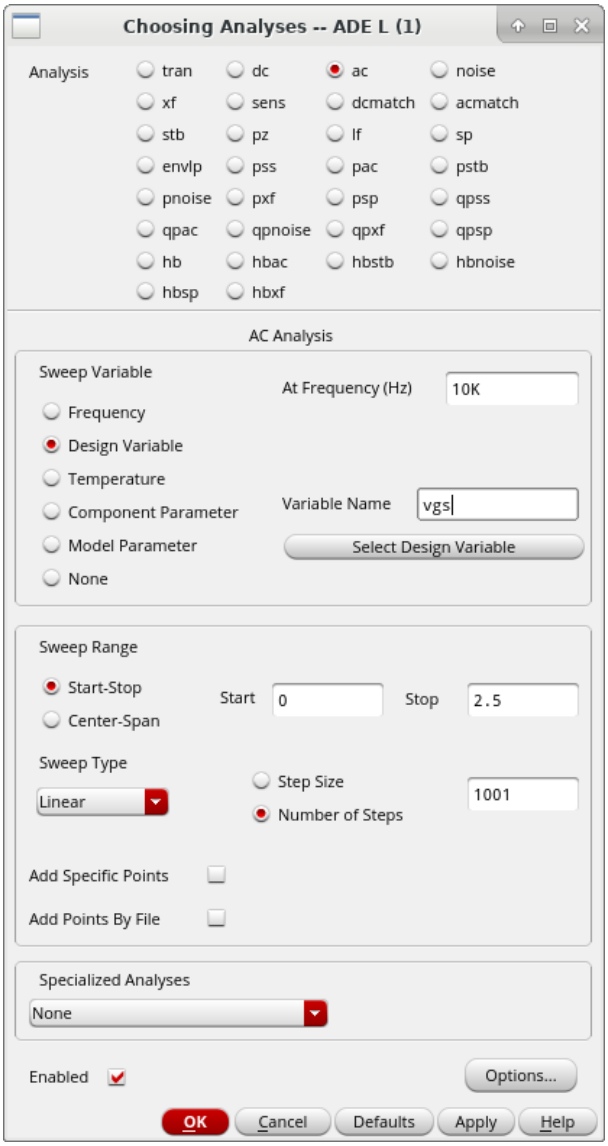
(5.1.4) In the schematic editor window, modify the gate-to-source DC voltage source (**vdc**) of the NMOS FET transconductance device to include an AC stimulus. (**Menu: Edit à Properties à Objects**, or **Edit Properties** icon-button, or press '**Q**'.) In the edit object properties form, set the 'AC magnitude' parameter to 1. Leave all other attributes unchanged. OK the form.



< Figure 70. Adding a 1V AC Stimulus to the DC Gate-to-Source Voltage >

(5.1.5) From the ADE window, setup the AC sweep analysis for I_{DS} vs. V_{GS} (**Menu: Analyses à Choose**). We wish to sweep the input voltage (V_{GS}) of the NMOS device from 0V to 2.5V while observing the drain current (I_{DS}). We will use 1001 steps in the sweep to give the output waveform sufficient resolution. We will not be sweeping the frequency since (for this exercise) we are only interested in the DC operating point behaviour of the AC analysis, so we will fix the frequency at 10KHz – at this relatively low frequency we can ignore the effects of the parasitic capacitances.

In the choosing analyses form, select the '**ac**' option button. Select the 'Design Variable' option as the sweep variable. Type *vgs* as the variable name. Set the 'At Frequency (Hz)' to 10K. Choose the 'Start-Stop' option button with Start = 0 and Stop = 2.5 for the sweep range. Select 'Linear' from the drop down menu as the sweep type. Choose the 'Number of Steps' option with 1001 steps in the sweep. Ensure that the analysis is enabled by checking the 'Enabled' box near the bottom of the form. See Figure 14. OK the form. The **ac** analysis should appear in the 'Analyses' list of the ADE window (next to the **dc** analysis we set up previously).



< Figure 71. The Choosing Analyses Form for a AC sweep. >

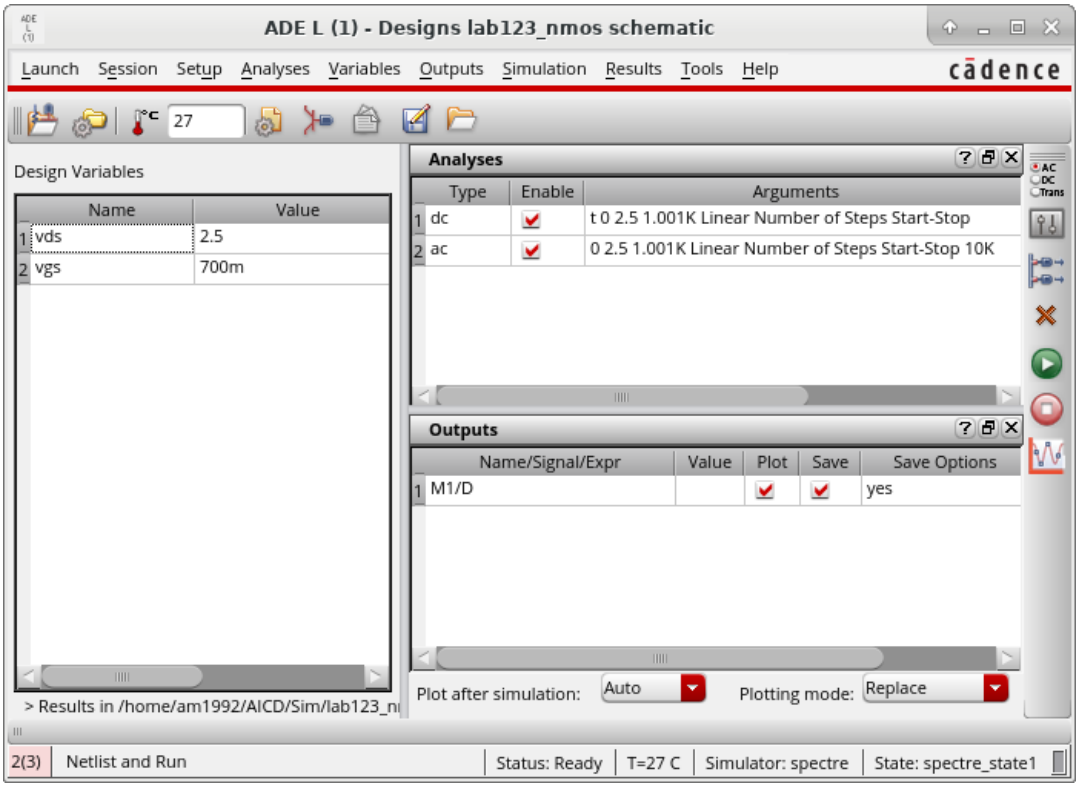
(5.1.6) If not already set up (from the last part), select the drain node current (I_{DS}) for plotting. From the ADE, select **Menu: Outputs à To Be Plotted à Select On Schematic** to enter the plotting mode of the ADE**.** Select only the drain node of the NMOS device (M1/D). **Press the 'ESC' key to exit the plotting mode of the ADE**.

(5.1.8) In the ADE window, run the '**Netlist and Run**' command by clicking on the green play button. The simulation should run without error and both the text output window as well as the waveform output window should open automatically. You will see two plots, in separate sub-windows. One will be a plot of I_{DS} vs. V_{GS} for the DC analysis, and one will be a plot of I_{DS} vs. V_{GS} for the AC analysis. See Figure 16.

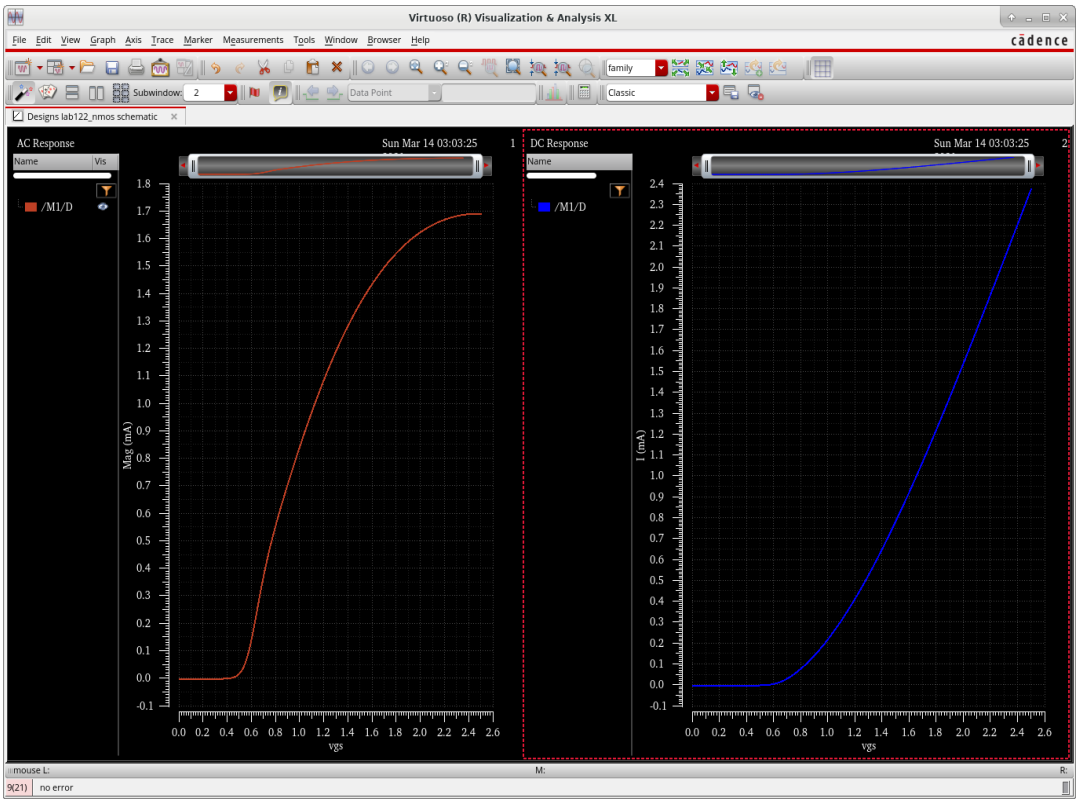
(5.1.9) Click on the DC analysis plot sub-window (note the dashed red line around the selected sub-window). Plot the curve of dI_{DS}/dV_{GS} vs. V_{GS} using the calculator just as you did previously i.e., deriv(IS("/M1/D")). Make sure the plotting mode of the calculator is set to 'append'. When you evaluate the buffer, the curve of dI_{DS}/dV_{GS} vs. V_{GS} will be added to the DC analysis plot sub window. See Figure 18.

(5.1.10) Note, the plot of dI_{DS}/dV_{GS} vs. V_{GS} in the DC analysis plot sub-window is the same as the g_m vs. V_{GS} curve in the AC analysis plot sub-window. Select the plot from the DC analysis plot sub-window and drag it over to the AC analysis plot sub-window. They will overlay *almost* exactly (the difference is miniscule). Whereas the AC analysis is implicitly calculating the g_m operating point *at* each sample of V_{GS}, the calculator derived plot from the DC analysis is calculating the differential *between* samples. So there may be a very small difference. (In fact if you use more steps per sample you see the difference between the AC and DC derived plots of gm disappear.

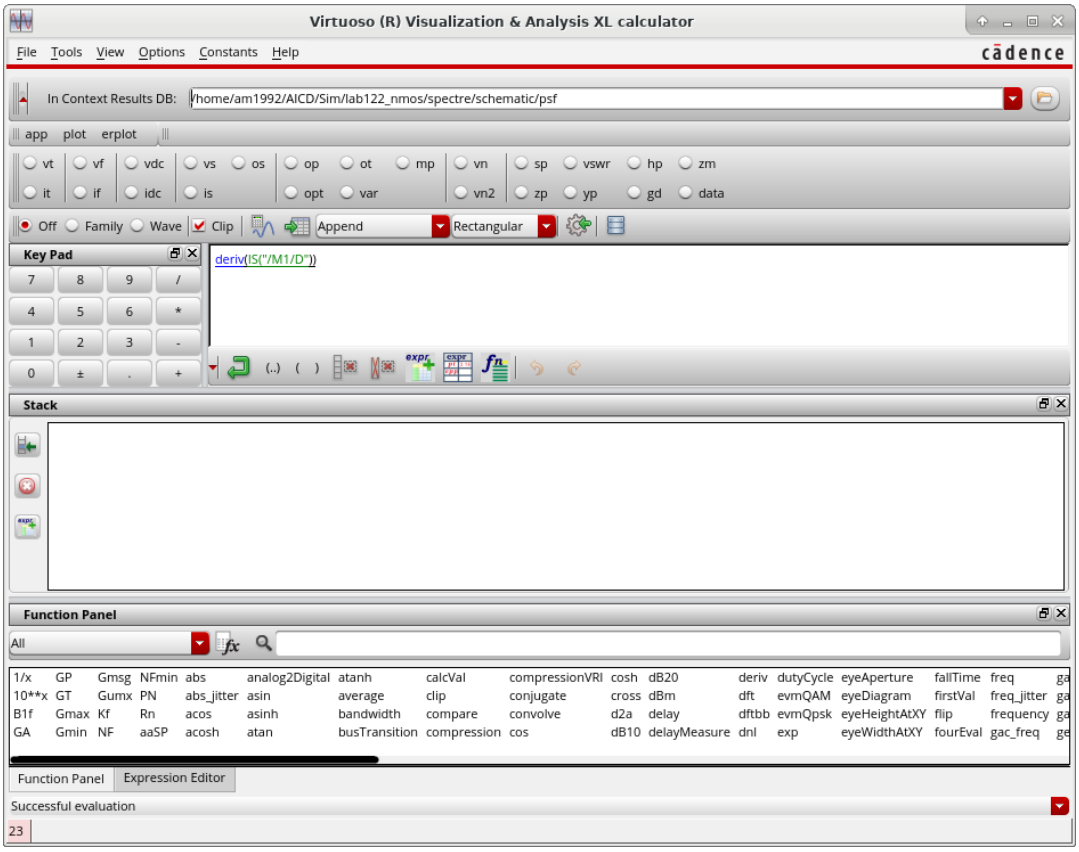
(5.1.11) In the final part of this lab, we will carry out a frequency sweep of the basic single-stage resistor-loaded NMOS FET amplifier. A frequency sweep calculates the output response to a steady-state sinusoidal input excitation over a range of frequencies.



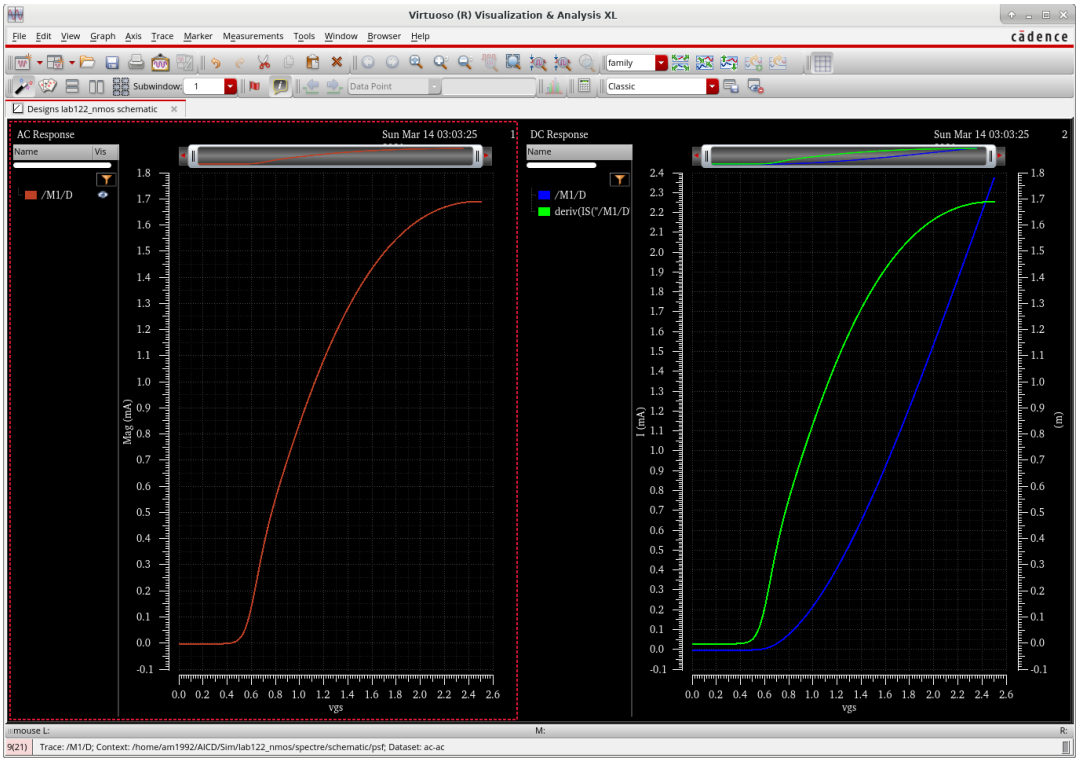
< Figure 72. The ADE with the Design Variables, Analyses and Outputs. >



< Figure 73. AC and DC Sweep of I_{DS} vs. V_{GS}. >



< Figure 74. The calculator set up to plot dI_{DS}/dV_{GS}. >



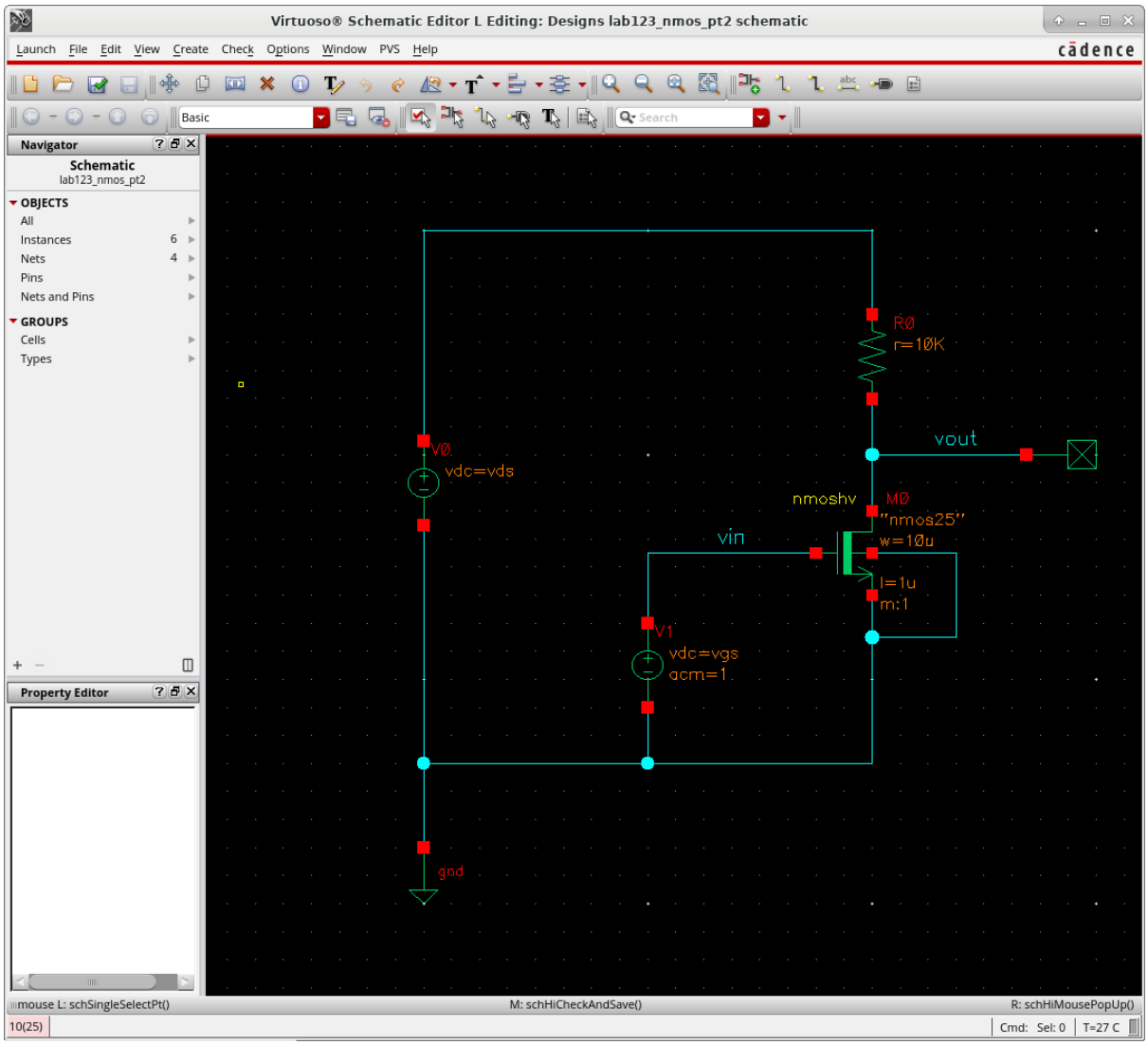
< Figure 75. AC and DC Sweep of I_{DS} vs. V_{GS}, and dI_{DS}/dV_{GS} vs. V_{GS} >

5.2 AC Analysis: Frequency Response

(5.2.1) Create a schematic of a resistor-loaded single-stage NMOS FET. See Figure 19.

(5.2.2) You can reuse your circuit from last lab (lab122_nmos_pt2 or whatever name you chose), you can copy the circuit from last lab, or you can create a new schematic from scratch. If you make a copy/new schematic, give it a meaningful name e.g., lab123_nmos_pt2.

Ensure that your schematic is properly parameterised using design variables: for the DC supply voltage vdc=vds and for the DC gate-to-source voltage vdc=vgs. The gate-to-source input voltage *vin* should have a 1V AC stimulus (AC magnitude = 1). The NMOS device should be W/L = 10u/1u. The resistor load should be 10KΩ. Name the output net *vout*. Use the noConn part to avoid floating net warnings. Ensure that it checks and saves without warnings/errors. Use the CIW to diagnose any warning or errors.

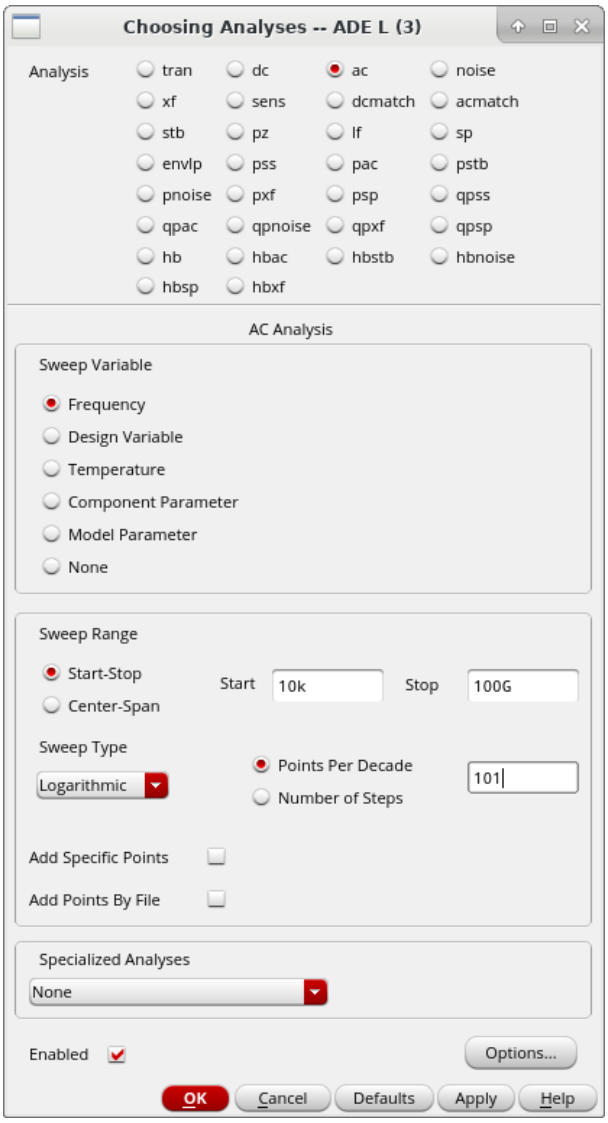


< Figure 76. Basic Resistor-Loaded Amplifier including a 1V AC Stimulus for AC Analysis>

(5.2.3) From the schematic window, launch the ADE tool (**Menu: Launch à ADE L**). From the ADE window, copy the design variables from the schematic into the ADE (**Menu: Variables à Copy From Cellview**). Initialise *vds = 2.5* and *vgs = 0.7*.

(5.2.4) From the ADE window, setup the AC sweep analysis (**Menu: Analyses à Choose**). In this simulation we will sweep the frequency from 10KHz to 100GHz. We will plot the output *vout* vs. frequency. Because of the large frequency range, we will use a logarithmic sweep of the frequency – with 101 plot points per decade of frequency.

In the choosing analyses form, select the '**ac**' option button. Select the 'Frequency' option as the sweep variable. Choose the 'Start-Stop' option button with Start = 10K and Stop = 100G for the sweep range. Select 'Logarithmic' from the drop down menu as the sweep type. Choose the 'Points Per Decade' option button with 101 plot points per decade. Ensure that the analysis is enabled by checking the 'Enabled' box near the bottom of the form. See Figure 20. OK the form. The **ac** analysis should appear in the 'Analyses' list of the ADE window (next to the **dc** analysis we set up previously).



< Figure 77. Choosing Analyses Form set up for a Frequency Sweep AC Analysis >

(5.2.5) Select the output net voltage (vout) for plotting. In the Analog Design Environment window, select **Menu: Outputs à To Be Plotted à Select On Schematic** to enter the plotting mode of the ADE**.** The schematic window will automatically become current. Select only the output net *vout*. **Press the 'ESC' key to exit the plotting mode of the ADE**.

(5.2.6) In the ADE window, run the '**Netlist and Run**' command by clicking on the green play button. The simulation should run without error and both the text output window as well as the waveform output window should open automatically. See Figure 22.

(5.2.7) Note that because we set *vin* = 1V, the plot of *vout* is the plot of the circuit voltage gain A_V. That is,

$$v_{out} = A_V v_{in} = A_V$$

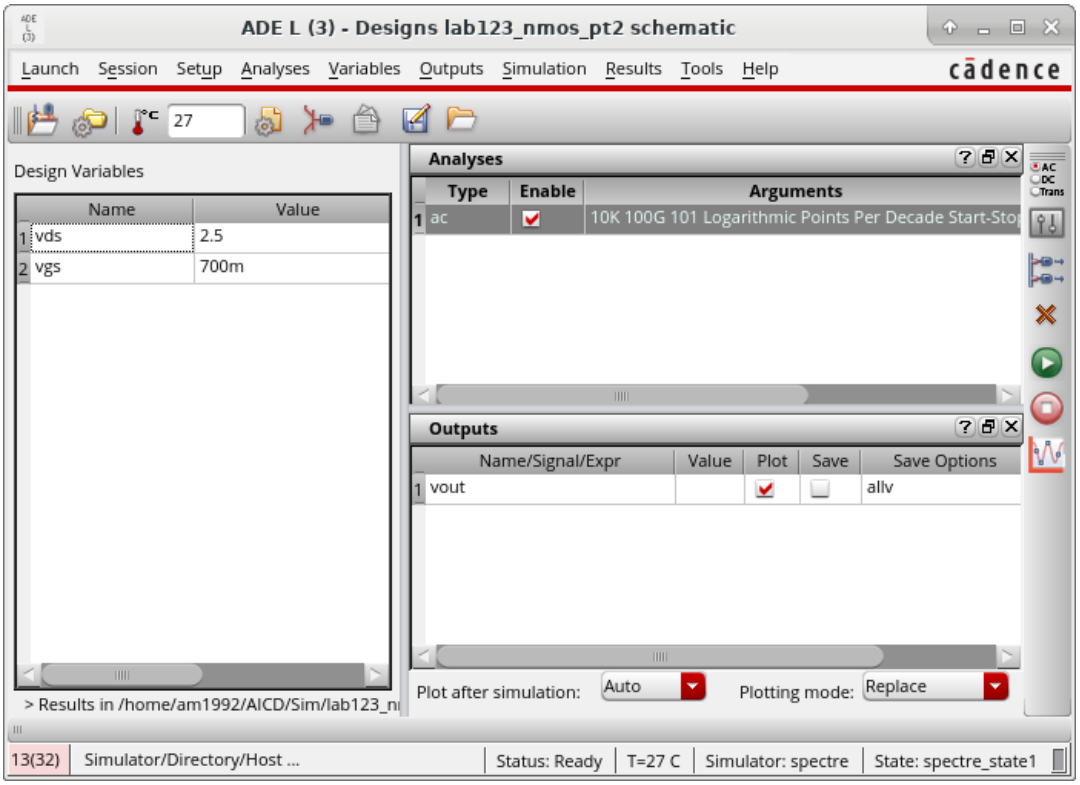
(5.2.8) The amplitude of the response by default is plotted on the linear/numerical scale – aka a **magnitude plot**. It is common practice to plot the output response of an amplifier in decibels (dB) to give a **log-log Bode** plot. (We use the dB scale as it makes it easier to graphically add responses, we can do this using the calculator.) From the ADE window, choose **Menu: Tools à Calculator…**

(5.2.9) Use the schematic Selection Toolbar to get the plot of *vout* into the calculator. Select the '**vf**' option button (for frequency sweep) on the schematic selection toolbar. When you select the '**vf**' button the schematic editor window will come into focus. Select only the output net *vout*. The calculator should return to focus, and you should see the expression for the drain current VF("/vout") in the buffer area of the calculator. (You may need to delete any expressions in the buffer from the previous exercises.)

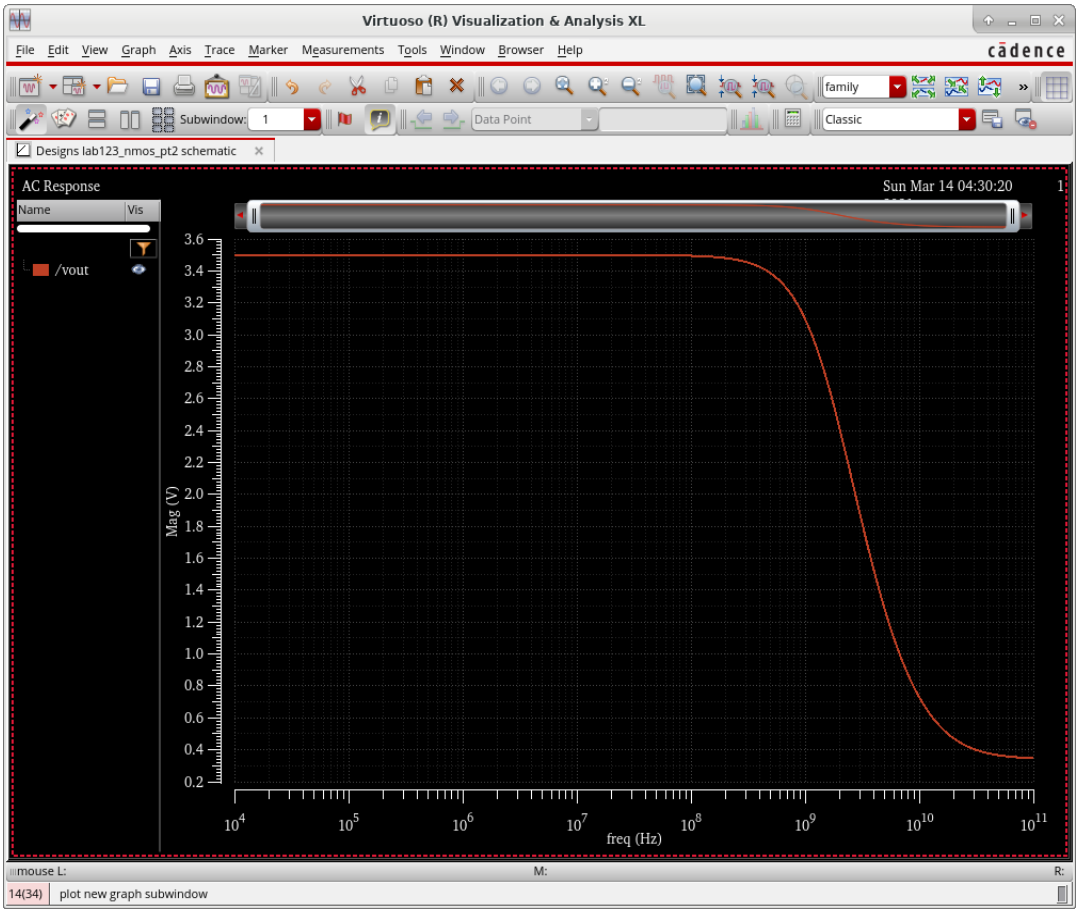
(5.2.10) With the expression for *vout* in the buffer, apply the decibel function from the function panel. In the function panel of the calculator window, select '**All**' from the drop down menu. Place the cursor in the correct positon in the buffer and select the '**dB20**' function from the list of functions. The functions are listed alphabetically so either search the list or use the text search box just above the listing area. Make sure to close the brackets after you select the **dB20** function. The expression in the buffer should be dB20(VF("/vout")). See Figure 24.

Note that there are two dB functions: dB10 and dB20. When converting a power gain *G* to decibels we use dB10 (that is $dB_{10} = 10 \log G$) and for a current/voltage gain *A* we use dB20 (that is $dB_{20} = 20 \log A$).

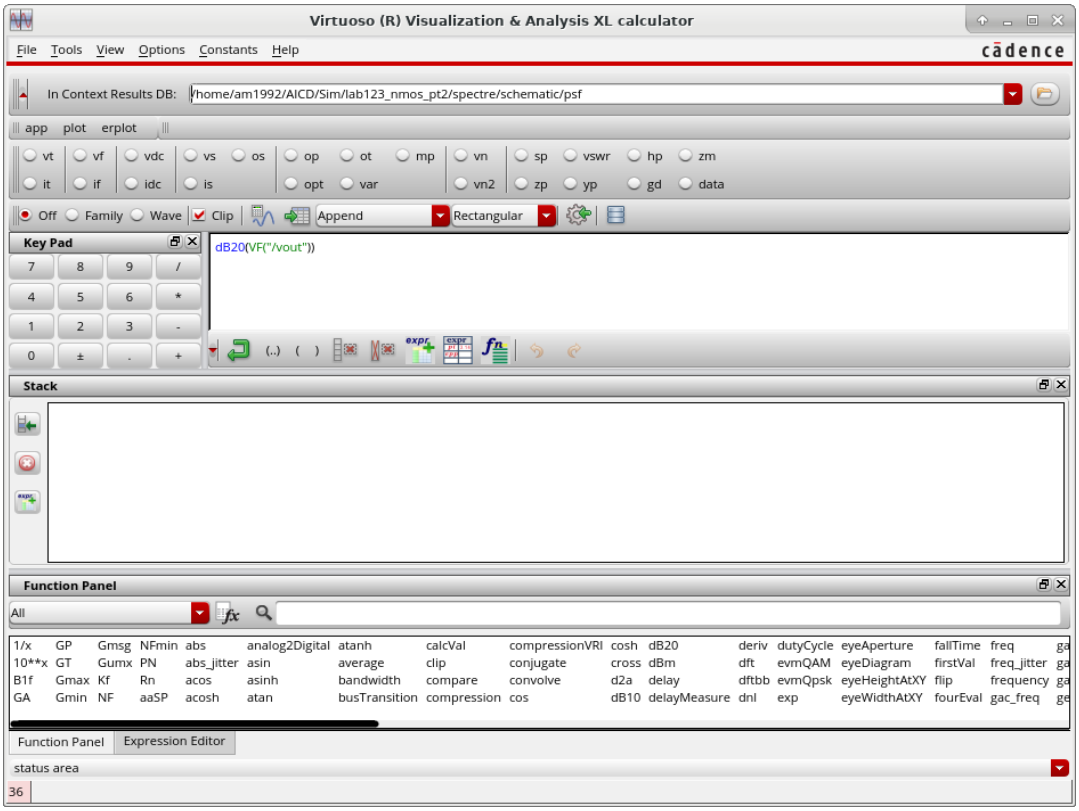
(5.2.11) In the calculator window, evaluate the buffer expression by using the **Evaluate the Buffer** button. If you set up the expression correctly and the simulation data exists (you should have run the AC sweep simulation before opening the calculator) the plot of $20 \log vout$ should be plotted alongside the original plot of vout.



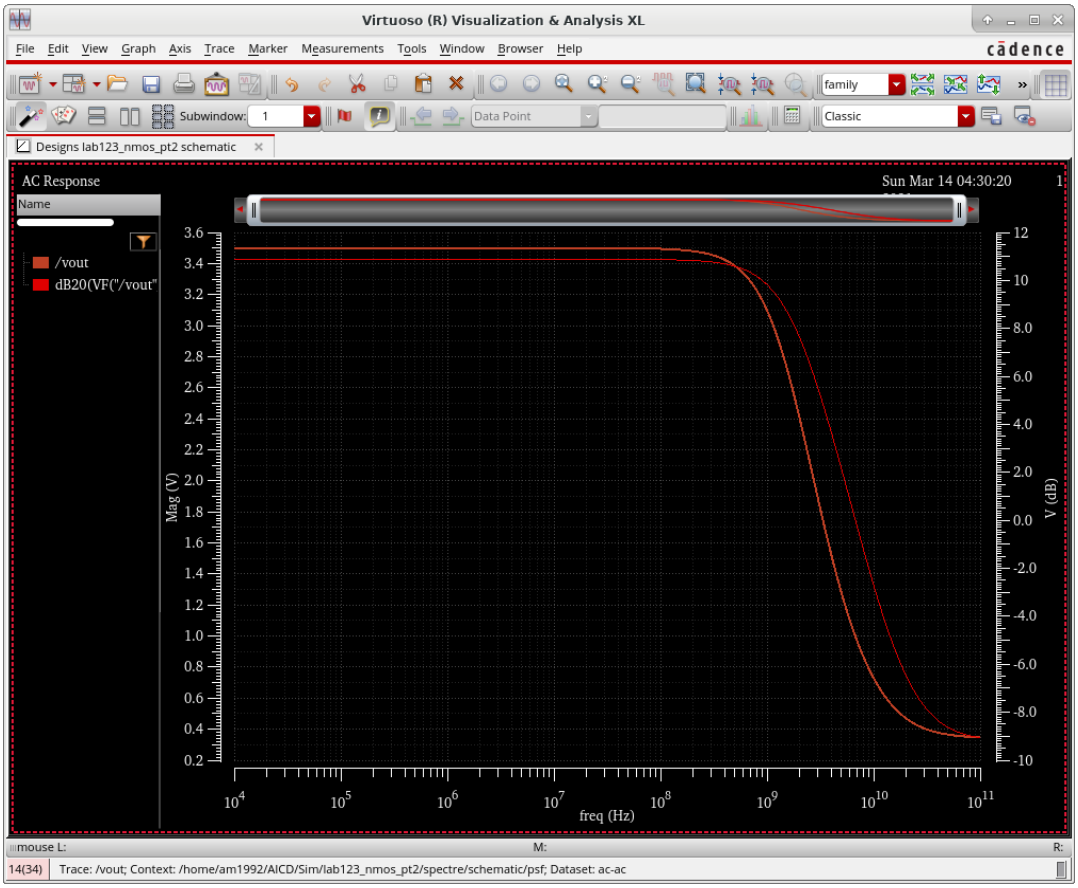
< Figure 78. The ADE with the Design Variables, Analyses and Outputs. >



< Figure 79. Frequency sweep response of the Resistor-Loaded NMOS FET Amplifier >



< Figure 80. Calculator setup to plot vout in dB (20 log vout) >



< Figure 81. Frequency response with Magnitude and dB Scales >

(5.2.12) We will discuss the frequency response of the MOSFET in more detail in the coming labs. Here, we will point out that the frequency sweep response of Figures 22 and 24 are generated from the linearised model of the MOSFET. We note that the gain of the MOSFET falls by 3dB between 1 to 2GHz. We can therefore assume that at least one pole exists around this point – where $\omega_{-3dB} = 1/RC$ – caused by the internal parasitic capacitances/resistances within the MOSFET. And at least one zero exists beyond 10GHz responsible for the flattening out of the response.

We will investigate the role of poles and zeros in more detail – particularly in the context of negative feedback – in the coming labs.

End of Lab!